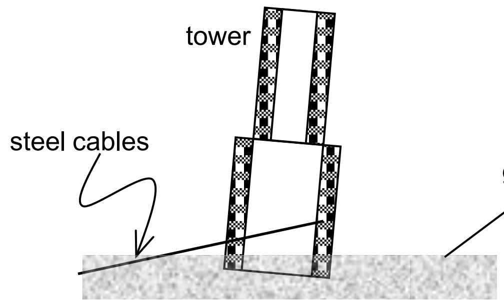
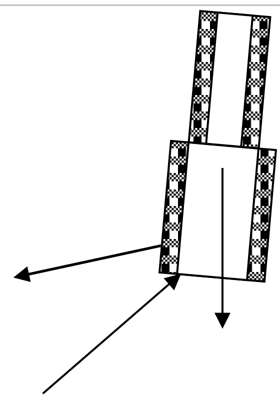
6 A tower is pulled by a bundle of steel cables anchored to the ground, as shown by the simplified sketch below.

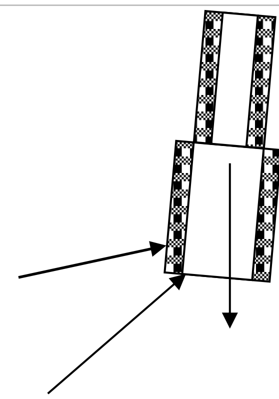


The tower is acted upon by three forces. Which diagram could show the position and direction of each of the forces when the tower is in equilibrium?

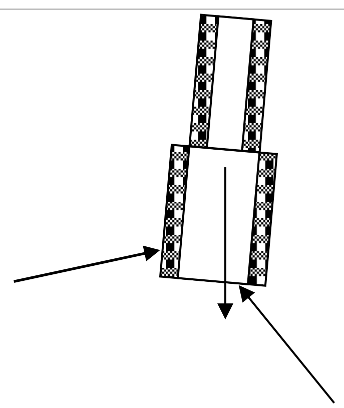
A



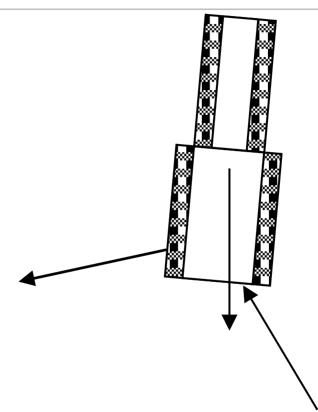
B



C



D



Answer: A

If 3 coplanar and non-parallel forces act on a body which is in equilibrium, the lines of action of all 3 forces must pass through a common point. C and D are eliminated.

Option B is incorrect as the steel cable is in tension not compression and there is a net force to the right and therefore the tower would not be in equilibrium.

7	A diving board of length 5.0 m is hinged at one end and supported 2.0 m from this end by a